

Unit - 5

Ergonomics

- Ergonomics is an applied scientific discipline that is concerned with how humans interact with the tools & equipment they use while performing tasks and other activities.
- Common focus of ergonomics is the interface between humans and the equipments.
- Work activities impose a continuum of physical and cognitive demands on the human worker i.e. most activities require a combination of physical & cognitive exertions.
- Ergonomics is also concerned with the physical and social environment in which the tasks & activities are performed and how humans and machines interact with the environment.

Ergonomics:
 Ergon: work
 omics: rules/laws
Study of work

- A professional whose practice is primarily focused on ergonomics is called as ergonomist.

Most work activities requires a combination of physical & cognitive

pure physical Requirements: > muscle strength > physical conditioning	pure cognitive Requirements: > Intelligence > Concentration > Knowledge
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→ Overview of Ergonomics

> Objectives and applications of Ergonomics:

- The main objective of ergonomics is to improve the performance of systems consisting of people and equipment. These systems are referred to as human-machine systems.

- Better performance of human-machine systems in an ergonomics content means the following:

- * Greater ease of interaction b/w the user & the equipment
- * Reduced stress and fatigue while using the equipment
- * Avoidance of errors and mistakes by the user.
- * Greater comfort and satisfaction in use of the equipment
- * Greater efficiency & productivity
- * Safer operation of the equipment
- * Avoidance of accident and injuries

- Ergonomics has a wide range of applications, which can be classified into two main areas:

* Work System design:

This area is concerned with the interaction b/w worker and the equipment used in the workplace. Objectives include safety, accident avoidance and related performance. This WSD includes consideration of factors related to work environment such as lighting and noise levels.

* Product Design:

This area deals with the ^{design} of products that are safer, more useful and more user friendly & mistake proof. In addition to providing greater customer satisfaction by means of these kinds of products, an issue in product design is product liability lawsuits. Their avoidance through consideration of ergonomics.

> Fitting the Job to the person

A common employment practice prior to ergonomics was based on philosophy called "fitting the person to the job" (FPJ), which recommended that workers be selected on the basis of their mental aptitudes and physical characteristics for a particular job opening.

- The FPJ approach is still considered among the eligibility factors for certain positions in many hiring situations today.
- Educational qualification is also and psychometric testing (ex: testing for intelligence and personality characteristics) are the parameters of FPJ approach.
- The ergonomics approach is diametrically opposite of FPJ. The philosophy in ergonomics is designing the job so that nearly any member of the workforce can perform it.
- Why ergonomics did not or superseded the FPJ approach:
 - (1) changes in worker skill requirements
 - (2) Demographic changes.
 - (3) social and political changes.

→ Human-Machine Systems

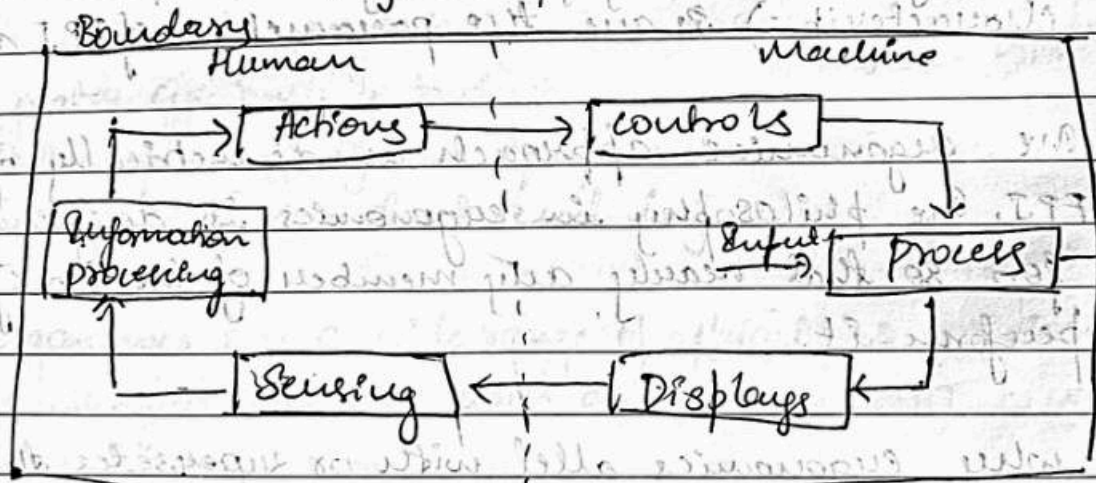
Defined as a combination of humans and equipment interacting to achieve some desired result.

- Human-m/c systems can be classified into 3 categories:
 - * Manual Systems: This system involves a person using

some hand tool or other non powered implement to perform an activity.

- * **Mechanical systems:** This system refers to one or more human using powered equipment to accomplish some job. The typical case is that the equipment provides the mechanical power to the job, & the fun of the human is to control the equipment.
- * **Automated Systems:** This system involves the performance of a job with a minimum of human attention for making repairs, reprogramming or loading & unloading of parts from the production m/c and to design & install automated.

- **Block diagram model of the interactions in human-m/c & physical & social environment**



- * **Boundary:**
A human-m/c system has boundaries that define what components are included within the scope of the system of analysis and design.

- * **Human components:**

The human components of interests are those that perform these functions:

(1) **Sensing (Human Senses):** The five basic senses are vision, hearing, touch, taste and smell. They are the sensors by which humans are able to be aware

of where they are and what is happening around them.

(2) Information processing (human brain): The human brain can be considered as a data and information processing unit, analogous in some ways to the operation of a computer. It receives the data through the human senses as well as feedback from the joints & muscles of the body. It stores the data in memory. It operates on the data by thinking, planning, calculating, solving problems and making decisions about what actions to take. These thought processes include both logical & cognitive activities. The decisions are communicated to and executed by the human effectors.

(3) Actions (Human effectors): An effector is a body part such as a muscle or group of muscles that actuates in response to some stimulus. The principal human effectors in the human-m/c system are the fingers, hands, feet & voice. These are supported by the musculoskeletal system of the body & stimulus is provided by the information processing occurring in human brain.

* Machine components:

The m/c components in the human-m/c system can range from a simple hand tool to a complex sophisticated system of equipment.

The three principle components of the m/c in human machine system:

(1) The process: The process refers to the function or operation performed by the human-m/c system.

(2) Displays: For processes that are simple & the worker participates ~~directly~~ directly, the display funⁿ is accomplished by the worker observing the process as it is being performed. No additional display of information is needed. For processes that are more mechanized and automated, there is a greater need for artificial displays to allow the human to see &

understand what is going on. High automated process like the chemical process require the worker to obtain the necessary information entirely from the display ~~monitors~~ monitor because the process itself cannot be directly observed.

(3) Controls: The human interacts with the equipment using m/c controls that can be operated by the human effectors usually the fingers and hands. When the equipment is simple, the control consist of one of the ~~stand~~ equipment components. As the machine becomes more complex, the controls tend to be more remote from the actual operation performed by human-m/c system.

* Environmental components:

A human-m/c operates in an environment that includes two components:

(a) Physical Environment: The physical environment includes the immediate area of the human-m/c system, separated from the system by a defined boundary. Components of physical environment usually include the location, surrounding lighting, noise, temp & humidity. These environmental factors can affect the performance of the human-m/c system, are of interest to the ergonomist.

(b) Social Environment: Work serves a social function & is a chance for workers to meet & interact with other people.

→ Topic Areas in Ergonomics

• Physical Ergonomics:

Many work activities require physical exertion. Physical ergonomics is mostly about manual labour. It is concerned with how the human body functions during physical exertion and how physical dimensions of the body affect the capabilities of the worker.

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monitors

* physiology: physiology is the branch of biology that deals with the vital processes carried out by living organisms and how their constituent tissues and cells function. Our main interest is directed at how the human muscles function during work activities. The energy expenditure that is required in the activity must be in balance with the capacity of the body's cardiovascular & respiratory systems. Rest periods must be allowed for work periods involving strenuous physical output.

* Anthropometry: It concerned with the physical dimensions & other ^{data} related to human body, such as height, reach & weight. The data for a given dimension is normally distributed for a defined population, but there are differences in these dimensions between defined populations. One widely applied principle is "design for extreme individuals", which recommends that the designed entity should accommodate virtually all of the user population.

* Cognitive Ergonomics: It is concerned with the capabilities and limitations of the human brain & sensory system while performing activities that include a significant amount of information processing. The human cognitive processes include the following components:

* Sensory System: This system consists of the five senses, that are activated by external stimuli such as sights & sounds.

* Perception: This is the stage of cognition that follows the sensing of external stimuli. It occurs when the human mind becomes aware of the sensation & interprets it based on previous experience and knowledge.

* Memory: Human memory consists of two main components: working memory (temporary memory) & long-term memory.

* Response selection & execution: This is the cognitive process of

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figuring out what actions are needed in light of information perceived through the sensory system and processed through working memory and long-term memory.

• The Physical Environment:

The Physical Environment can be classified into the following components:

* The visual environment: is what the worker sees while working. Most of the information input perceived by the worker is through visual stimuli. The important factor enabling the proper functioning of a worker's sense of vision is lighting. The term that describes how well things can be seen is visibility. The job of the ergonomist is to determine the optimum illumination levels for the task & to implement other design details in the workplace in order to achieve the best possible visual environment.

* Auditory environment: is what the worker hears while working. Included in the auditory environment is noise, which can be defined as sound that is undesired & possibly harmful to the worker's sense of hearing. The most important factors about noise in terms of its effects on humans are: (1) loudness (2) duration of exposure.

* Climate: is what worker feels while working. Climate is defined by (1) air temperature (2) humidity (3) air movement (4) radiation from surrounding objects. However, the climatic conditions that cause heat stress and cold stress often require interventions beyond the body's natural ~~process~~ responses. The interventions include proper clothing for the conditions, frequent rest breaks and heating and air conditioning of the facility.



→ Occupational Safety and Health

- Every year in the U.S., approx. 5000 lives are lost due to occupational injuries & 50,000 lives are lost due to work related diseases and illnesses.
- The following federal agencies were established with the enactment of the Occupational Safety and Health Act in 1971:
 - * Occupational Safety and Health Administration (OSHA), whose responsibilities include enforcing the provisions of OSHA Act, implementing safety and health programs & setting occupational safety and health standards.
 - * National Institute for Occupational Safety & Health (NIOSH), whose responsibility is to engage in research, training & education in the area of occupational safety and health.
- Occupational safety is concerned with the avoidance of industrial accidents, which are one time events that can cause injury or fatality.
- Occupational health is concerned with avoiding diseases and disorders that are caused by workers being exposed to hazardous materials or conditions in the workplace. These diseases & disorders do not generally occur immediately upon exposure. Instead they tend to develop after prolonged periods of exposure to the materials or conditions. The effects of the exposure are cumulative & may take years before symptoms reveal the onset of the malady.